

Coding & Solution

Date	30 September 2026
Team ID	[Your Team ID]
Project Name	AI BlogNest
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements for the AI BlogNest backend.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub repository link for AI BlogNest here]
Programming Language(s)	JavaScript (Node.js)
Framework(s) Used	Express.js, node-postgres (pg), @google/generative-ai (Gemini SDK)
Key Features Implemented	User registration & login (JWT + bcrypt.js), authenticated profile access, full blog CRUD (create/read/update/delete), AI blog generation via Gemini, AI blog summarization via Gemini, centralized error handling, request validation, CORS and request logging
Pending / Incomplete Features	Role-based access control (Admin/Editor/Reader), comment system, pagination on blog listing, email verification on signup
Setup / Run Instructions	1) npm install 2) configure .env with DATABASE_URL, JWT_SECRET, GEMINI_API_KEY 3) create PostgreSQL database "blognest" and run the users/blogs table SQL 4) npm run dev 5) server starts at http://localhost:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The backend was migrated from an original MongoDB/Mongoose prototype to PostgreSQL for stronger relational integrity between users and blogs (foreign key with ON DELETE CASCADE). All protected routes are guarded by JWT authentication middleware, and Gemini AI calls are isolated in a dedicated service layer for maintainability.
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